

Algebra II with Trigonometry

Chapter 6.2

Study guide

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[4506907_alg-2trig-h-chp-6-2-note-docx.pdf](#)

Certainly! Here is a complete, self-contained study guide based on the provided document.

Trigonometry Functions of Real Numbers - Study Guide

1. Short Summary

This document is a lesson in Algebra 2 Trigonometry studying the trigonometric functions as functions of real numbers. It introduces definitions of the six main trigonometric functions using the unit circle, states their domains, and explains how to determine their values and signs depending on the quadrant. It also covers key identities (reciprocal and Pythagorean), the even/odd properties of these functions, and provides step-by-step example problems for evaluation, applications, and compound functions.

2. Core Definitions, Theorems, and Formulas

Trigonometric Functions on the Unit Circle

For a real number t , let $P(x, y)$ be the terminal point on the unit circle (radius 1) determined by t :

- $\sin t = y$

- $\cos t = x$
- $\tan t = \frac{y}{x}$ (if $x \neq 0$)
- $\csc t = \frac{1}{y}$ (if $y \neq 0$)
- $\sec t = \frac{1}{x}$ (if $x \neq 0$)
- $\cot t = \frac{x}{y}$ (if $y \neq 0$)

Domains

| Function | Domain | |-----|-----| | sin, cos | all real numbers
 | | tan, sec | all real numbers except $\frac{\pi}{2} + n\pi$, n integer | | cot, csc | all real numbers except $n\pi$, n integer |

Signs in Quadrants

| Quadrant | Positive Functions | Negative Functions | |-----|-----|-----|
 -----| | I | all | None | | II | sin, csc | cos, sec, tan, cot | | III | tan, cot | sin, csc, cos, sec | | IV | cos, sec | sin, csc, tan, cot |

Reciprocal Identities

- $\csc t = \frac{1}{\sin t}$
- $\sec t = \frac{1}{\cos t}$
- $\cot t = \frac{1}{\tan t}$
- $\tan t = \frac{\sin t}{\cos t}$
- $\cot t = \frac{\cos t}{\sin t}$

Pythagorean Identities

- $\sin^2 t + \cos^2 t = 1$
- $\tan^2 t + 1 = \sec^2 t$
- $1 + \cot^2 t = \csc^2 t$

Even-Odd Properties

- *Odd functions:* $f(-x) = -f(x)$ — sine, cosecant, tangent, cotangent
- *Even functions:* $f(-x) = f(x)$ — cosine, secant

- $\sin(-t) = -\sin t$, $\cos(-t) = \cos t$, $\tan(-t) = -\tan t$
 - $\csc(-t) = -\csc t$, $\sec(-t) = \sec t$, $\cot(-t) = -\cot t$
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3. Step-by-Step Study Guide

A. Understanding Trigonometric Functions on the Unit Circle

1. The unit circle is the circle with center at the origin (0,0) and radius 1. 2. A real number t represents the length of arc from the point (1,0) counterclockwise. 3. The coordinates (x, y) of the corresponding point define:

- $\cos t = x$
- $\sin t = y$
- The other functions are defined as ratios involving x and y .

B. Determining Signs of Functions

- **Quadrants:**
- I (0 to $\pi/2$): All functions are positive.
- II ($\pi/2$ to π): only $\sin$, $\csc$ positive.
- III (π to $3\pi/2$): only $\tan$, $\cot$ positive.
- IV ($3\pi/2$ to 2π): only $\cos$, $\sec$ positive.

C. How to Find a Trigonometric Value for Any Real Number

1. **Find the reference number** (reference angle): It's the acute angle between the terminal side of t and the x-axis. 2. **Determine the sign**: Use the quadrant to decide if the function is positive or negative. 3. **Calculate the value**: Use known values for the reference angle and apply the appropriate sign.

D. Using the Fundamental Identities

- **Reciprocal and Pythagorean identities** are crucial for rewriting and evaluating expressions, especially when you know one value and need to find another.

E. Even and Odd Properties

- Use these to simplify expressions involving negative angles.

F. Example Problem Types

- Find exact values given an angle, sometimes outside $[0, 2\pi]$.
 - Find the values of all trig functions at given coordinates.
 - Determine signs and quadrants based on trig function properties.
 - Rewrite functions in terms of others using identities.
 - Apply trigonometric functions to real-life models, such as oscillation.
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4. Practice Problems with Answers

1. Evaluate $\cos\left(-\frac{\pi}{4}\right)$.

Answer: $\cos\left(-\frac{\pi}{4}\right) = \cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$ (cosine is even).

2. What is the sign of $\tan t$ if t is in Quadrant IV?

Answer: Negative (only cos and sec are positive in Quadrant IV).

3. Find $\sec t$ if $\cos t = -\frac{3}{5}$ and t is in Quadrant III.

Answer: $\sec t = \frac{1}{\cos t} = -\frac{5}{3}$

4. If $\sin t = \frac{4}{5}$ and t is in Quadrant II, find $\cos t$.

Answer: Use $\sin^2 t + \cos^2 t = 1$:

$$\cos^2 t = 1 - \left(\frac{4}{5}\right)^2 = 1 - \frac{16}{25} = \frac{9}{25}. \text{ Since Quadrant II, } \cos t < 0:$$
$$\cos t = -\frac{3}{5}$$

5. Write $\tan t$ in terms of $\sin t$ and $\cos t$.

Answer: $\tan t = \frac{\sin t}{\cos t}$

5. Assumptions and Ambiguities

- The exact methods for finding reference numbers or angles were assumed to be known, though not explicitly defined.
- Some notations were inferred from context (e.g., the convention that t is the angle in radians).
- It was assumed the user is comfortable with unit circle values and radian measure.
- Not all answer steps for example questions were shown in the document, so detailed work for practice problem answers was supplied.
- For applications (like Example 9), the formula for displacement was used as given, without context on solving the full table.

Let me know if you want worked solutions to the in-document examples or practice problems!